Two permanent congruences for shifted difference matrices

Principia Math in Progress

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Task 1. Prove both permanent congruences. Let $D_N = [i-j]_{1\le i,j\le N}$ and $B_N = I_N + D_N$. Wang and Sun [1], p. 3, conjecture

$$\operatorname{per} B_{p-1} \equiv 3 \pmod p, \qquad \operatorname{per} B_p \equiv 1 + 4p \pmod{p^2}$$

for odd primes p. MathDB also includes $p = 2$ in the first assertion; that case is immediate from $B_1 = (1)$. We prove the stronger polynomial statements below. Congruences of polynomials mean coefficientwise congruences.

Theorem 1. *For every odd prime p,*

$$\operatorname{per}(I_{p-1} + tD_{p-1}) \equiv 1 + 2t^{p-1} \pmod p, \qquad \operatorname{per}(I_p + tD_p) \equiv 1 + 4pt^{p-1} \pmod{p^2}.$$

We first prove a symmetric-polynomial fact. For variables $x_1, \dots, x_N$, write

$$\operatorname{per}[\delta_{ij} + t(x_i - x_j)] = \sum_{d=0}^{N} c_d(x_1, \dots, x_N) t^d, \qquad P_j = \sum_{i=1}^{N} x_i^j.$$

The polynomial c_d is symmetric and homogeneous of degree d. Also $c_0 = 1$, and $c_d = 0$ for odd d, since transposition replaces t by $-t$.

Lemma 1. *For $1 \le 2m \le N$, the expression of c_{2m} in the power sums has the form*

$$c_{2m} = -(m-1)!N^{\underline{m}}P_{2m} + \textit{terms containing at least two positive-index power sums},$$

where $N^{\underline{m}} = N(N-1)\cdots(N-m+1)$. Every coefficient in this expression has denominator divisible only by primes at most $2m$.

Proof. The denominator assertion follows from the fundamental theorem of symmetric polynomials and Newton's identities: an integral symmetric polynomial of degree $2m$ is integral in the elementary symmetric polynomials, and converting those of degree at most $2m$ to power sums introduces only the integers $1, \dots, 2m$ as divisors. There is no constant term by homogeneity.

Here is an explicit calculation of the coefficient of P_{2m}. Define a linear functional $\mathcal{W}$ on monomials in four auxiliary variables by

$$\mathcal{W}(u^a r^b v^c s^d) = \begin{cases} a!c!, & a = b,\ c = d, \\ 0, & \text{otherwise.} \end{cases}$$

Expansion of the rank-two factorization $t(x_i - x_j) = (tx_i, 1)(1, -tx_j)^T$ gives

$$\operatorname{per}[\delta_{ij} + t(x_i - x_j)] = \mathcal{W} \prod_{i=1}^{N} \bigl(1 + vr + tx_i(ur - vs) - t^2 x_i^2 us\bigr). \tag{1}$$

Indeed, selecting a subset of the factors selects a principal permanent; the factorials count all matchings between the two types of chosen row and column terms.

Put

$$\ell_j = [z^j] \log\bigl(1 + vr + z(ur - vs) - z^2 us\bigr), \quad j \ge 1.$$

The product in (1) is

$$(1 + vr)^N \exp\left(\sum_{j\ge 1} t^j P_j \ell_j\right).$$

Consequently the coefficient of P_{2m} in c_{2m} is $C_{2m}(N) = \mathcal{W}((1+vr)^N \ell_{2m})$. These are formal coefficient calculations: only finitely many auxiliary monomials can survive $\mathcal{W}$ at any fixed power of t, so no analytic convergence is involved.

For clarity, expand the logarithm and let q be the number of quadratic terms $-z^2us$ chosen. The remaining $2m-2q$ terms come from zur and $-zvs$. The functional forces us to take exactly q factors vr from $(1+vr)^{N-(2m-q)}$. Summing over the two linear choices gives

$$\sum_{a+b=2m-2q} (-1)^b \frac{(a+q)!}{a!}\frac{(b+q)!}{b!} = (q!)^2[z^{2m-2q}](1-z^2)^{-q-1} = (q!)^2\binom{m}{q}.$$

Thus

$$C_{2m}(N) = -\sum_{q=0}^{m}(2m-q-1)!q!\binom{m}{q}\binom{N-2m+q}{q}. \tag{2}$$

The finite Chu–Vandermonde identity evaluates this sum as

$$C_{2m}(N) = -(2m-1)!\frac{(-N)_m}{(1-2m)_m} = -(m-1)!N^{\underline{m}},$$

where $(a)_m = a(a+1)\cdots(a+m-1)$. For example, after dividing the sum in (2) by $-(2m-1)!$, its summand is $\frac{(-m)_q(N-2m+1)_q}{(1-2m)_q q!}$, the terminating binomial identity. None of its denominator factors is zero. This proves the lemma. □

Proof of Theorem 1. First let $N = p-1$ and reduce modulo p. Simultaneously permuting the row and column indices by multiplication by any $a \in \mathbb{F}_p^\times$ replaces t by at. Hence the polynomial $\operatorname{per}(I + tD_N)$, of degree at most $p-1$, has no terms except degrees 0 and $p-1$. Its constant term is one. To calculate the top coefficient, for arbitrary x_i the rank-two expansion gives

$$\operatorname{per}[x_i - x_j] = \sum_{k=0}^{N}(-1)^{N-k}k!(N-k)!e_k(x)e_{N-k}(x),$$

where e_k denotes the elementary symmetric polynomial. For x_i running through $\mathbb{F}_p^\times$, $\prod_i(X-x_i) = X^{p-1}-1$. Thus only $k = 0, N$ remain, $e_N = -1$, and Wilson's theorem gives $N! = -1$. Since N is even, the two terms each equal one. This proves the first polynomial congruence.

Now set $N = p$, $x_i = i$, and work modulo p^2. The usual finite-field power sums give

$$P_j \equiv 0 \pmod p \quad (1 \le j < p-1), \qquad P_{p-1} \equiv -1 \pmod p.$$

For $1 \le 2m \le p-1$, all denominators in Lemma 1 are invertible modulo p^2. Every term with at least two power sums vanishes modulo p^2: each of its indices is less than $p-1$, and each factor is divisible by p. Therefore

$$c_{2m}(1,\dots,p) \equiv -(m-1)!p^{\underline{m}}P_{2m} \pmod{p^2}. \tag{3}$$

If $2m < p-1$, both $p^{\underline{m}}$ and P_{2m} are divisible by p, so this coefficient is zero modulo p^2. For $m = (p-1)/2$, we have

$$-(m-1)!p^{\underline{m}} \equiv (-1)^m p\big((m-1)!\big)^2 \pmod{p^2}.$$

Wilson's theorem, paired around the midpoint, says $(m!)^2 \equiv (-1)^{m+1} \pmod p$; also $m^2 \equiv 1/4 \pmod p$. It follows that

$$(-1)^m\big((m-1)!\big)^2 \equiv -4 \pmod p.$$

Using $P_{p-1} \equiv -1 \pmod p$ in (3) gives $c_{p-1} \equiv 4p \pmod{p^2}$. The odd coefficients, including degree p, vanish identically. Together with $c_0 = 1$, this proves the second polynomial congruence. Specializing $t = 1$ proves both original assertions. □

Verification. The general proof above applies to every required prime. The accompanying script checks the finite coefficient identity symbolically, isolates its linear power-sum coefficient using finite-field roots of unity, and compares complete permanent polynomials with the two polynomial congruences for small primes. Independent direct permutation sums also check the permanent algorithm at small orders.

Task 2. Lift the congruence. Determine the complete polynomial $\operatorname{per}(I_p + tD_p)$ modulo p^3, including the contributions from products of two power sums.

Task 3. Vary the matrix order. Determine coefficientwise congruences modulo p^2 for $\operatorname{per}(I_{kp} + tD_{kp})$ when $k \geq 2$, accounting for the higher-degree terms whose power-sum expansions can have denominators divisible by p.

Task 4. Extend the kernel. Find analogous congruences for low-rank antisymmetric polynomial kernels $K(x, y) = -K(y, x)$.

Task 5. Evaluate exact permanents. Derive a closed formula or a short recurrence for $\operatorname{per}(I_N + tD_N)$ at arbitrary integer orders N.

References

[1] H. Wang and Z.-W. Sun, *Evaluations of some Toeplitz-type determinants*, arXiv:2206.12317v8, 14 February 2023. The two conjectures appear together on p. 3 of the actual 22-page PDF.